SMART INDIA HACKATHON 2025

SMART INDIA HACKATHON 2025

SP

- **Problem Statement ID –** SIH25033
- **Problem Statement Title–** AI Based Smart Allocation Engine for PM Internship Scheme
- **Theme-** Smart Automation
- **Category-** Software
- **Team ID-** 67524
- **Team Name (Registered on portal) -** SP

SP

AI Based Smart Allocation Engine for PM internship scheme

SMART INDIA HACKATHON 2025

❖ Proposed Solution

Overview -

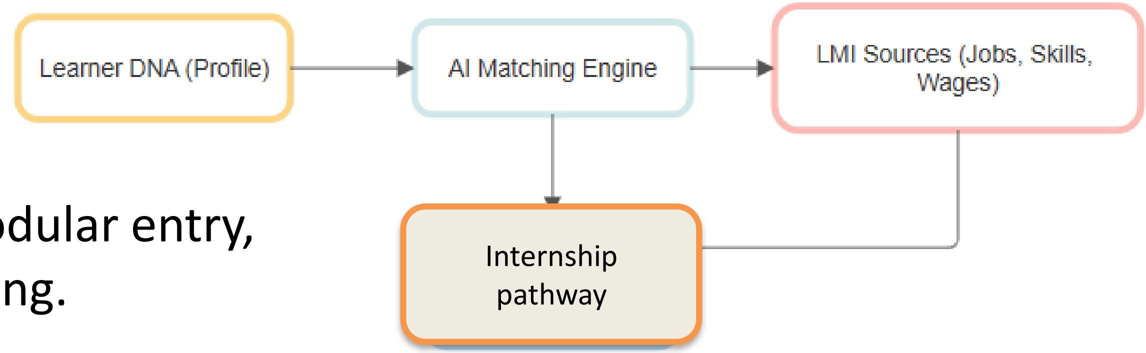
Opportunity Intelligence: Aggregate industry internship data, government programs, regional trends, and skill projections for optimal placement.

DNA Profiling of Applicants: Capture personal, educational, skills, certification, and aspirational data for targeted internship matching.

How it addresses the problem -

Mismatch Prevention: Prevents poor allocation by aligning every recommendation to current opportunity demand and certified competencies, improving placement rates.

Candidate Support: Enables lifelong upskilling with micro-credentials, stackable modules, and pathways for modular entry, allowing flexible participation and progress without restarting.



Innovation and Differentiation –

Real-time Industry Alignment: Dynamically links candidates to current internships, projects, and government opportunities based on progress.

Future-Oriented Guidance: Leverages market analysis to suggest internships and roles likely to be in demand in the next 2–5 years.

SP

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025

❖ Technology stack

Core AI/ML

AI matchmaking engine using transformer-based NLP for candidate-role fit, preferences parsing, and dynamic scoring.

Embeddings, vector databases, and RAG (retrieval-augmented generation) for intelligent matching and contextual recommendations.

Data and knowledge

Candidate profiling database (relational/NoSQL) storing skills, education, certifications, and internship preferences.

Internship opportunity database aggregating roles, requirements, locations, partner organization data, and market trends.

Industry & Policy Alignment

Sync with national internship databases (e.g., NSQF/NCVET) for standards, credit systems, and compliant outcome reporting.

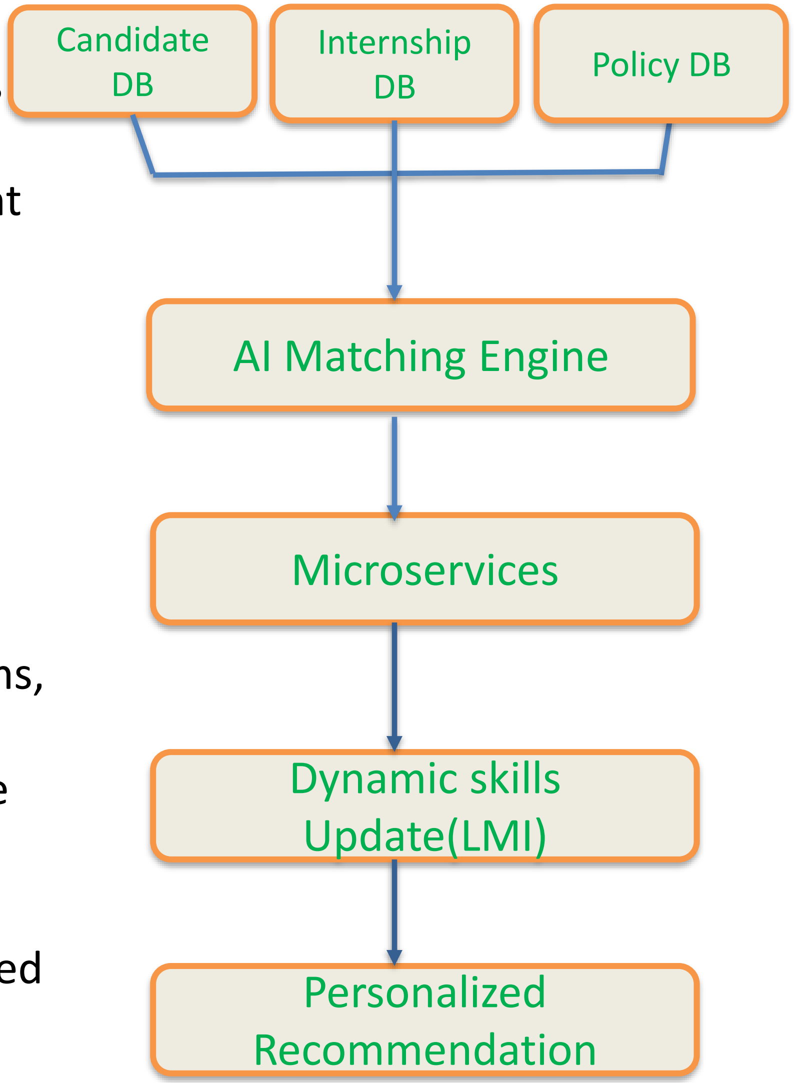
Map competency frameworks and credit transfer schemes for verified skills and module progress.

Scalable architecture

Microservices for matchmaking inference, notification dispatch, and data sync—deployed via Node.js or Python/Flask APIs.

Real-time Labour Market Intelligence

ML normalization of skill terms and detection of emerging skills for dynamic path updates.



SP

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Feasibility

- Technical: Proven stack with RAG (embeddings + vector DB), LLM orchestration, and skills/qualifications graph; cloud-native microservices enable autoscaling and low-latency APIs.

Visibility

- Scalable operations: Automated ingestion, multilingual UX, WCAG compliance, monitoring, and retraining loops support sustained expansion.

Challenges

- Data quality/availability: Heterogeneous NSQF/NOS formats, incomplete course metadata, and noisy LMI streams.
- Model reliability/bias: Hallucinations, multilingual variance, and difficulty misclassification impacting trust.

Strategies

- Data governance: Standard schemas, validation pipelines, human-in-the-loop checks for mappings and high-stakes content.
- Guardrails: Retrieval-augmented generation, constrained decoding, confidence scoring, explainability, and multilingual evaluations.

USE CASE :-

- Personalized NSQF-aligned learning paths that adapt milestones, difficulty, and modality based on diagnostics and prior learning.
- Real-time job-aligned recommendations where LMI signals re-rank courses and suggest emerging skills, internships, and apprenticeships.
- Employer pipelines that verify micro-credentials and readiness, with auto-suggested bridge modules to reach role requirements.
- Policymaker dashboards to align funding with regional skill demand, track outcomes, and audit qualification recognition.

SP

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

❖ Potential impact on the target audience

- Increased internship placement rates by matching candidates to opportunities best aligned to their skills, preferences, and career goals, improving job readiness and future prospects.
- Greater accessibility for students from diverse educational, regional, and socio-economic backgrounds, ensuring equitable opportunity distribution.
- Enhanced support for underrepresented groups and persons with disabilities through compliant and localized platform features.
- Real-time analytics empower institutions to identify placement gaps, monitor candidate progress, and intervene proactively.

❖ Benefits of the solution

- Social: Broadens reach and inclusivity, connecting underserved groups with competitive internships through transparent and unbiased AI matching.
- Educational: Accelerates practical skill acquisition and workforce integration by aligning opportunities with curriculum and recognized learning paths.
- Economic: Optimizes utilization of government and industry resources, boosting employability and reducing transition time from study to work.
- Technological: Ensures scalable, secure, and adaptive placement processes with modern AI-driven matching, analytics, and data security protocols.
- Environmental: Digital application and selection workflow reduces paperwork, travel, and physical resource consumption, promoting sustainability.

SP

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

❖ Personalized learning and planning

Retrieval-augmented generation in education: survey and components for grounding, retrieval, and generation.

- ❖ Practitioner guides and tools for RAG stacks in 2025, including orchestration and retrieval choices.
- ❖ Case projects on AI-powered learning path generation and RAG-based recommendation prototypes.
- ❖ Generative AI for personalized tutoring and feature-conditioned prompts for adaptive guidance.

❖ NSQF/NCVET alignment

- ❖ NSQF overview and level descriptors used to align competencies and outcomes.
- ❖ National Credit/Qualification framework notification and descriptors for mapping credits and outcomes.
- ❖ NCVET role in qualification approval, monitoring, and recognition across providers.
- ❖ Example NSQF Qualification File structure for program alignment and assessment standards.